Lecture 13

The characters of S_n

Objectives. *Lecture 12 labelled the irreducibles of S_n by partitions. This lecture finds all their characters at once. A class function becomes a symmetric polynomial, its* Frobenius characteristic*; the irreducible characters become the* Schur functions *s_λ, quotients of two determinants; and one coefficient gives any value $\chi^\lambda(\mu)$.*

By the end of this lecture you will be able to:

- Prove that the characteristic is an isometry $\mathrm{CF}(S_n) \to \Lambda^n$ that turns restriction to S_{n-1} ($n \geq 2$) into $\partial/\partial p_1$.
- Prove that the s_λ, $\lambda \vdash n$, are an orthonormal basis of Λ^n, with $\partial s_\lambda/\partial p_1 = \sum_{\mu=\lambda-\square} s_\mu$ for $n \geq 2$.
- Prove that $\mathrm{ch}(\chi^\lambda) = s_\lambda$ for every $\lambda \vdash n$.
- Compute any value $\chi^\lambda(\mu)$ by the Frobenius formula.

13.1 Symmetric functions and the characteristic

Fix $n \geq 1$ and $N \geq n$ variables $x_1, \ldots, x_N$. For $\alpha \in \mathbb{Z}^N_{\geq 0}$, write $x^\alpha = x_1^{\alpha_1} \cdots x_N^{\alpha_N}$, and $[x^\alpha]f$ for the coefficient of x^α in a polynomial f. A partition λ of length at most N is padded with zeros to length N.

Definition 13.1 (Symmetric polynomials)**.** Λ^n is the space of polynomials f in $x_1, \ldots, x_N$ with complex coefficients, homogeneous of degree n and *symmetric*: $f(x_{\sigma(1)}, \ldots, x_{\sigma(N)}) = f(x_1, \ldots, x_N)$ for every $\sigma \in S_N$. For $\lambda \vdash n$, the *monomial* m_λ is the sum of the distinct x^α with α a rearrangement of λ.

A symmetric f has the same coefficient at x^α as at its rearrangement x^λ, so

$$f = \sum_{\lambda \vdash n} [x^\lambda] f \; m_\lambda,$$

and the $p(n)$ monomials m_λ, $\lambda \vdash n$, are a basis of Λ^n, as $\ell(\lambda) \leq n \leq N$. Setting $x_{N+1} = 0$ sends each m_λ in $N+1$ variables to m_λ in N: an isomorphism, which also keeps the p_μ and h_μ below. So the choice of $N \geq n$ does not matter.

Definition 13.2 (Power sums and complete sums)**.** For $k \geq 1$,

$$p_k = x_1^k + \cdots + x_N^k,$$

and h_k is the sum of all monomials of degree k. For a partition $\mu = (\mu_1, \ldots, \mu_\ell)$, $p_\mu = p_{\mu_1} \cdots p_{\mu_\ell}$ and $h_\mu = h_{\mu_1} \cdots h_{\mu_\ell}$.

Subscripts keep p_k, h_k apart from $p(n)$ and group elements.

Lemma 13.3 (The power-sum basis)**.** *The p_μ, $\mu \vdash n$, are a basis of Λ^n.*

Proof. Expanding $p_\mu = \prod_j (\sum_i x_i^{\mu_j})$, each monomial puts every part μ_j on one variable. Its exponents, sorted, are μ, or, if two parts share a variable, a partition of smaller length. And x^μ occurs $\prod_i m_i(\mu)!$ times, once for each way to send the parts to $x_1, \dots, x_\ell$ as x^μ needs:

$$p_\mu = \Big(\prod_i m_i(\mu)!\Big) m_\mu + \big(\text{terms } m_\lambda,\ \ell(\lambda) < \ell(\mu)\big).$$

List the partitions of n by decreasing length: the matrix taking the m_λ to the p_μ is triangular with non-zero diagonal. ∎

Definition 13.4 (The Hall product and the characteristic)**.** The *Hall product* on Λ^n is

$$\langle p_\lambda, p_\mu \rangle = z_\lambda \, \delta_{\lambda\mu},$$

extended antilinearly in the first slot and linearly in the second, like $\langle \cdot, \cdot \rangle_{S_n}$. The *Frobenius characteristic* of $\chi \in \mathrm{CF}(S_n)$ is

$$\begin{aligned} \mathrm{ch}(\chi) &= \sum_{\mu \vdash n} \chi(\mu) \, \frac{p_\mu}{z_\mu} \\ &= \frac{1}{n!} \sum_{\sigma \in S_n} \chi(\sigma) \, p_{\mu(\sigma)}, \end{aligned}$$

where $\chi(\mu)$ is the value on the class of type μ, of size $n!/z_\mu$ (Theorem 2.9), and $\mu(\sigma)$ is the type of σ.

Differentiating $p_\mu = p_1^{m_1(\mu)} p_2^{m_2(\mu)} \cdots$ in p_1 defines a linear map $\partial/\partial p_1 \colon \Lambda^n \to \Lambda^{n-1}$ on this basis ($\Lambda^0 = \mathbb{C}$):

$$\frac{\partial p_\mu}{\partial p_1} = m_1(\mu) \, p_{\mu^-},$$

where μ^- is μ with one part 1 removed (0 if $m_1(\mu) = 0$).

Theorem 13.5 (The characteristic)**.** *Let $n \geq 1$ and $N \geq n$.*

(i) $\mathrm{ch} \colon \mathrm{CF}(S_n) \to \Lambda^n$ *is a linear isomorphism, and* $\langle \mathrm{ch}\, \varphi, \mathrm{ch}\, \psi \rangle = \langle \varphi, \psi \rangle_{S_n}$.

(ii) For $\chi \in \mathrm{CF}(S_n)$ and $\mu \vdash n$,

$$[x^\mu] \, \mathrm{ch}(\chi) = \langle \chi_{M^\mu}, \chi \rangle_{S_n}.$$

So for a representation V, $\mathrm{ch}(\chi_V)$ has non-negative integer coefficients; and

$$\mathrm{ch}(1_{S_n}) = h_n = \sum_{\mu \vdash n} \frac{p_\mu}{z_\mu}.$$

(iii) If $n \geq 2$, then $\mathrm{ch}\big(\mathrm{Res}^{S_n}_{S_{n-1}} \chi\big) = \partial \, \mathrm{ch}(\chi)/\partial p_1$.

Proof. (i) ch takes the indicators of the classes, a basis of $\mathrm{CF}(S_n)$, to the basis p_μ/z_μ of Λ^n (Lemma 13.3). And

$$\begin{aligned} \langle \mathrm{ch}\, \varphi, \mathrm{ch}\, \psi \rangle &= \sum_\mu \overline{\varphi(\mu)} \, \psi(\mu) \, \frac{z_\mu}{z_\mu^2} \\ &= \frac{1}{n!} \sum_\mu \frac{n!}{z_\mu} \, \overline{\varphi(\mu)} \, \psi(\mu) \\ &= \langle \varphi, \psi \rangle_{S_n}, \end{aligned}$$

by the class form of $\langle\cdot,\cdot\rangle_{S_n}$ (Lecture 6).

(ii) *The count.* Let $\nu = \mu(\sigma)$. Then

$$p_\nu = \prod_j \Big(\sum_i x_i^{\nu_j}\Big)$$

has one factor for each cycle of σ, so it is a sum over the ways F to give each cycle a variable, and F gives x^μ exactly when, for each i, the cycles sent to x_i cover a set A_i of μ_i points. So $[x^\mu]p_\nu$ counts the $(A_1,\dots,A_\ell) \in X_\mu$ with every A_i a union of cycles of σ: it is $\chi_{M^\mu}(\sigma)$ (Proposition 8.14(ii)).

The average. As χ_{M^μ} is real,

$$\begin{aligned}[x^\mu]\operatorname{ch}(\chi) &= \frac{1}{n!}\sum_\sigma \chi(\sigma)\,\chi_{M^\mu}(\sigma)\\ &= \langle \chi_{M^\mu}, \chi\rangle_{S_n}.\end{aligned}$$

For $\chi = \chi_V$ it is $\dim \operatorname{Hom}_{S_n}(M^\mu, V)$ (Theorem 6.7), and $\operatorname{ch}(\chi_V)$ is symmetric.

The trivial character. For $\chi = 1_{S_n}$ it is the number of orbits on X_μ, which is 1 (Example 5.8): every monomial of degree n has coefficient 1, as in h_n.

(iii) If $m_1(\mu) \geq 1$, then $z_\mu = m_1(\mu)\, z_{\mu^-}$, so

$$\frac{\partial}{\partial p_1}\frac{p_\mu}{z_\mu} = \frac{p_{\mu^-}}{z_{\mu^-}},$$

and the terms with $m_1(\mu) = 0$ vanish. Type ν in S_{n-1} is type $\nu \cup (1)$ in S_n, so

$$\begin{aligned}\frac{\partial \operatorname{ch}(\chi)}{\partial p_1} &= \sum_{\nu \vdash n-1} \chi\big(\nu \cup (1)\big)\frac{p_\nu}{z_\nu}\\ &= \operatorname{ch}\big(\operatorname{Res}^{S_n}_{S_{n-1}}\chi\big). \quad \blacksquare\end{aligned}$$

Example 13.6 (Guard: z_μ, not the class size)**.** For $n = 2$, $\operatorname{ch}(1_{S_2}) = (p_1^2 + p_2)/2$, and z_μ is 2 for both classes:

$$\big\langle \operatorname{ch} 1_{S_2}, \operatorname{ch} 1_{S_2}\big\rangle = \frac{1}{4}(2+2) = 1.$$

With the class sizes, 1 and 1, in place of the z_μ, it is

$$\frac{1}{4}(1+1) = \frac{1}{2}:$$

the isometry, and Cauchy's identity in § 13.2, need the z_μ.

13.2 Schur functions, Cauchy and $\partial/\partial p_1$

For $\alpha \in \mathbb{Z}^N_{\geq 0}$, the *alternant* is

$$a_\alpha = \det\big(x_j^{\alpha_i}\big)_{i,j=1}^N = \sum_{w\in S_N} \operatorname{sgn}(w)\prod_i x_{w(i)}^{\alpha_i},$$

and $\delta = (N-1, N-2, \dots, 0)$, the *staircase* (a Kronecker $\delta_{\lambda\mu}$ carries two indices). Exchanging two variables exchanges two columns: a_α is *antisymmetric*, changing sign. If two α_i are equal, $a_\alpha = 0$; and permuting the entries of α by u multiplies a_α by $\operatorname{sgn}(u)$.

Lemma 13.7 (Alternants).

(i) An antisymmetric polynomial f in $x_1, \dots, x_N$, with coefficients polynomials in other variables, is $\prod_{i<j}(x_i - x_j) \cdot g$ with g symmetric.

(ii) $a_\delta = \prod_{i<j}(x_i - x_j)$, of degree $N-1$ in each x_i.

Proof. (i) *One factor.* Dividing by $x_i - x_j$ as a polynomial in x_i leaves the remainder $f|_{x_i=x_j}$, which is 0: exchanging x_i and x_j fixes it and negates it.

All factors. Divide by the $x_i - x_j$ one at a time: at $x_i = x_j$ the earlier factors are non-zero, so the quotient so far vanishes there. The quotient g of two antisymmetric polynomials is symmetric.

(ii) By antisymmetric divisibility, a_δ is the product times a constant g, as both have degree $N(N-1)/2$. And $g = 1$, by induction on N: the coefficient of x_1^{N-1} is $a_\delta(x_2, \dots, x_N)$ in the determinant (along the first column) and $\prod_{2\le i<j}(x_i - x_j)$ in the product. ∎

Definition 13.8 (Schur functions). For a partition λ with $\ell(\lambda) \le N$, the *Schur function* is

$$s_\lambda = \frac{a_{\lambda+\delta}}{a_\delta},$$

a symmetric polynomial of degree $|\lambda|$, by Lemma 13.7.

Its subscript is a partition; a Coxeter generator s_i has an integer. Write $h_n(xy)$ for h_n in the N^2 variables $x_i y_j$.

Lemma 13.9 (Cauchy's identity). *In $x_1, \dots, x_N$ and $y_1, \dots, y_N$, for every $n \ge 0$,*

$$\sum_{\lambda \vdash n,\ \ell(\lambda)\le N} a_{\lambda+\delta}(x)\, a_{\lambda+\delta}(y) = a_\delta(x)\, a_\delta(y)\, h_n(xy).$$

Proof. Work with formal power series (infinite sums of monomials), with $1/(1-u) = \sum_k u^k$.

The determinant. By Leibniz,

$$\begin{aligned}\det\Big(\frac{1}{1-x_iy_j}\Big) &= \sum_w \operatorname{sgn}(w) \prod_i \sum_{k_i\ge 0} (x_i y_{w(i)})^{k_i} \\ &= \sum_{k\in\mathbb{Z}^N_{\ge 0}} x^k\, a_k(y).\end{aligned}$$

Here $a_k(y) = 0$ unless the k_i are distinct; the k that rearrange one $\lambda + \delta$ collect, with their signs, to $a_{\lambda+\delta}(x)\, a_{\lambda+\delta}(y)$.

Cauchy's determinant. Let $P = \prod_{i,j}(1 - x_i y_j)$. Then

$$P \det\Big(\frac{1}{1-x_iy_j}\Big) = \sum_w \operatorname{sgn}(w) \prod_i \prod_{j\ne w(i)} (1 - x_i y_j)$$

is a polynomial, antisymmetric in the x_i and in the y_j, of degree at most $N-1$ in each variable. By Lemma 13.7(i) in x, and comparing degrees in each x_i, it is $a_\delta(x)\, g$ with g free of x; by antisymmetric divisibility in y, $g = \kappa\, a_\delta(y)$ with constant κ.

The constant. Expanding $1/P$, every monomial in the $x_i y_j$ occurs once, so

$$\sum_\lambda a_{\lambda+\delta}(x)\, a_{\lambda+\delta}(y) = \kappa\, a_\delta(x)\, a_\delta(y) \sum_{n\ge 0} h_n(xy).$$

The terms of total degree $N(N-1)$ are the $\lambda = \varnothing$ term and $\kappa\, a_\delta(x)\, a_\delta(y)$, so $\kappa = 1$. The terms of total degree $N(N-1) + 2n$ give the lemma. ∎

Theorem 13.10 (Schur functions). *Let $n \geq 1$ and $N \geq n$.*

(i) The s_λ, $\lambda \vdash n$, are an orthonormal basis of Λ^n.

(ii) For $\mu \vdash n-1$, $p_1 s_\mu = \sum_{\lambda = \mu + \square} s_\lambda$; in particular $s_{(1)} = p_1$.

(iii) For $n \geq 2$ and $\lambda \vdash n$,

$$\frac{\partial s_\lambda}{\partial p_1} = \sum_{\mu = \lambda - \square} s_\mu .$$

Proof. (i) *Cauchy.* Divide Lemma 13.9 by $a_\delta(x)\, a_\delta(y)$, and apply Theorem 13.5(ii) in the $N^2 \geq n$ variables $x_i y_j$, where $p_k(xy) = p_k(x)\, p_k(y)$:

$$\begin{aligned} \sum_{\lambda \vdash n} s_\lambda(x)\, s_\lambda(y) &= h_n(xy) \\ &= \sum_{\rho \vdash n} \frac{p_\rho(x)\, p_\rho(y)}{z_\rho} . \end{aligned}$$

Coefficients. Write $s_\lambda = \sum_\rho b_{\lambda\rho}\, p_\rho$, with $b_{\lambda\rho}$ rational, as s_λ and the p_ρ are. The $p_\rho(x)\, p_\tau(y)$ are linearly independent, as bases in separate variables, so

$$\sum_\lambda b_{\lambda\rho}\, b_{\lambda\tau} = \frac{\delta_{\rho\tau}}{z_\rho} .$$

The real square matrix with entries $b_{\lambda\rho}\sqrt{z_\rho}$ has orthonormal columns, hence orthonormal rows:

$$\langle s_\lambda, s_\mu \rangle = \sum_\rho b_{\lambda\rho}\, b_{\mu\rho}\, z_\rho = \delta_{\lambda\mu} .$$

(ii) Multiplying the term of w in a_α by $x_{w(i)}$ raises α_i by 1, so

$$p_1 a_\alpha = \sum_{i=1}^{N} a_{\alpha + \varepsilon_i} ,$$

with ε_i the i-th unit vector. Take $\alpha = \mu + \delta$. If $\mu + \varepsilon_i$ is a partition, it is $\mu + \square$, and each $\mu + \square$ arises so, as $\ell(\mu + \square) \leq N$. Otherwise $\mu_{i-1} = \mu_i$, so $\alpha + \varepsilon_i$ has two equal entries, and $a_{\alpha+\varepsilon_i} = 0$. Divide by a_δ.

(iii) *Adjointness.* For $f \in \Lambda^{n-1}$ and $g \in \Lambda^n$,

$$\langle p_1 f, g \rangle = \langle f, \partial g / \partial p_1 \rangle .$$

On $f = p_\nu$ and $g = p_\mu$, both sides vanish unless $\mu = \nu \cup (1)$, and then they are z_μ and $m_1(\mu)\, z_\nu$, equal by the proof of Theorem 13.5(iii).

Expansion. Apply Schur orthonormality, adjointness and the one-box rule:

$$\begin{aligned} \frac{\partial s_\lambda}{\partial p_1} &= \sum_{\mu \vdash n-1} \langle s_\mu, \partial s_\lambda / \partial p_1 \rangle\, s_\mu \\ &= \sum_\mu \langle p_1 s_\mu, s_\lambda \rangle\, s_\mu \\ &= \sum_{\mu = \lambda - \square} s_\mu . \qquad \blacksquare \end{aligned}$$

13.3 Integrality and predecessors

Lemma 13.11 (Extraction). *For $f \in \Lambda^n$ and $\lambda \vdash n$ ($N \geq n$),*

$$\langle s_\lambda, f\rangle = [x^{\lambda+\delta}](a_\delta f).$$

Proof. Expand in the orthonormal basis (Theorem 13.10):

$$f = \sum_\nu \langle s_\nu, f\rangle\ s_\nu,$$
$$a_\delta f = \sum_\nu \langle s_\nu, f\rangle\ a_{\nu+\delta}.$$

The exponents of $a_{\nu+\delta}$ are the rearrangements of $\nu+\delta$, of which only $\nu+\delta$, with coefficient 1, is strictly decreasing, as $\lambda+\delta$ is. So $[x^{\lambda+\delta}]a_{\nu+\delta}$ is 1 if $\nu=\lambda$, and 0 otherwise. ∎

A *virtual character* of S_n is an integral combination of irreducible characters.

Corollary 13.12 (Integrality). *For $\lambda \vdash n$, $\mathrm{ch}^{-1}(s_\lambda)$ is a virtual character.*

Proof. Let $\varphi = \mathrm{ch}^{-1}(s_\lambda)$ and $U \in \mathrm{Irr}(S_n)$. By Theorem 13.5(i) and Lemma 13.11,

$$\langle \chi_U, \varphi\rangle_{S_n} = \langle \mathrm{ch}\,\chi_U, s_\lambda\rangle = \overline{[x^{\lambda+\delta}](a_\delta\,\mathrm{ch}\,\chi_U)},$$

an integer, as a_δ and $\mathrm{ch}\,\chi_U$ are integral (Theorem 13.5(ii)). The irreducible characters are an orthonormal basis of $\mathrm{CF}(S_n)$ (Theorem 7.9(ii)), so $\varphi = \sum_U \langle \chi_U, \varphi\rangle_{S_n}\,\chi_U$. ∎

Example 13.13 (Guard: norm one is not enough). Example 9.8's class function φ on S_3 has norm 1 and $\varphi(e) = 1$. But

$$\mathrm{ch}(\varphi) = \frac{1}{6}p_1^3 + \frac{1}{2}p_2p_1 - \frac{1}{3}p_3,$$

and

$$[x_1^3]\,\mathrm{ch}(\varphi) = \frac{1}{6} + \frac{1}{2} - \frac{1}{3} = \frac{1}{3},$$

which is $\langle 1_{S_3}, \varphi\rangle_{S_3}$: not an integer, so φ is not virtual (Theorem 13.5(ii)). Norm one and $\varphi(e) > 0$ make a character only with integrality (Example 6.10).

Lemma 13.14 (Predecessors). *For $n \geq 3$, a partition $\lambda \vdash n$ is determined by the set of partitions $\lambda - \square$. For $n = 2$, (2) and $(1,1)$ both have only (1).*

Proof. Let $\lambda \neq \lambda'$ have the same predecessors, among them μ. Then $\lambda = \mu + \square$ and $\lambda' = \mu + \square'$ with $\square \neq \square'$, so the diagrams meet in μ, of $n-1$ boxes.

Rectangles. Every predecessor of λ lies in both diagrams, so in μ, and is μ. With one removable box, λ is a rectangle (a^b); likewise $\lambda' = (c^d)$.

The count. As $\lambda \neq \lambda'$, $a \neq c$; say $a < c$. Then $b > d$, as ab and cd are both n. The diagrams meet in (a^d), so

$$ad = n - 1 = ab - 1.$$

Hence $a(b-d) = 1$:

$$a = 1, \qquad b = d + 1.$$

As cd is n, which is b, $d(c-1) = 1$:

$$d = 1, \qquad c = 2, \qquad n = 2.$$ ∎

13.4 The identification

Theorem 13.15 (The characters of S_n). *For $\lambda \vdash n \geq 1$ and $N \geq n$,*

$$\mathrm{ch}(\chi^\lambda) = s_\lambda.$$

Proof. Let $\varphi_\lambda = \mathrm{ch}^{-1}(s_\lambda)$.

The value at e. Pairing $p_{(1^n)}$ with $s_\lambda = \sum_\mu \varphi_\lambda(\mu)\, p_\mu/z_\mu$ gives $\varphi_\lambda(e) = \langle p_1^n, s_\lambda\rangle$. By adjointness and Theorem 13.10(iii), for $n \geq 2$,

$$\begin{aligned}\langle p_1^n, s_\lambda\rangle &= \langle p_1^{n-1}, \partial s_\lambda/\partial p_1\rangle \\ &= \sum_{\mu=\lambda-\square} \langle p_1^{n-1}, s_\mu\rangle,\end{aligned}$$

and $\langle p_1, s_{(1)}\rangle = 1$. So $\varphi_\lambda(e)$ counts the paths from (1) to λ in Young's graph: $f^\lambda \geq 1$ (Definition 12.1).

Irreducible. By Theorem 13.5(i) and Theorem 13.10(i),

$$\langle \varphi_\lambda, \varphi_\mu\rangle_{S_n} = \langle s_\lambda, s_\mu\rangle = \delta_{\lambda\mu}.$$

So φ_λ is virtual (Corollary 13.12), of norm 1 and positive at e: an irreducible character (Example 6.10).

Counting. There are $p(n)$ distinct φ_λ and $p(n)$ irreducibles (Theorem 12.5), so $\varphi_\lambda = \chi^{\pi(\lambda)}$ for a bijection π of partitions.

$n \leq 2$. $n = 1$ is clear. For $n = 2$, $\langle s_\lambda, p_2\rangle$ sums the coefficients of the $x^{\lambda+\delta-2\varepsilon_i}$ in a_δ (Lemma 13.11); only $i = 1$ gives a rearrangement of δ: δ for (2), δ with its first two entries exchanged for $(1,1)$. So the values $\overline{\varphi_\lambda((2))}$ are

$$\langle s_{(2)}, p_2\rangle = 1, \qquad \langle s_{(1,1)}, p_2\rangle = -1,$$

so

$$\varphi_{(2)} = 1_{S_2}, \qquad \varphi_{(1,1)} = \mathrm{sgn},$$

which are $\chi^{(2)}$ and $\chi^{(1,1)}$ (§ 12.2).

$n \geq 3$. With $\mathrm{ch}(\chi^\mu) = s_\mu$ for $\mu \vdash n-1$ (induction), Theorem 13.5(iii) and Theorem 13.10(iii) give

$$\begin{aligned}\mathrm{ch}\big(\mathrm{Res}^{S_n}_{S_{n-1}} \chi^{\pi(\lambda)}\big) &= \sum_{\mu=\lambda-\square} s_\mu \\ &= \mathrm{ch}\Big(\sum_{\mu=\lambda-\square} \chi^\mu\Big).\end{aligned}$$

By branching (Theorem 12.5(iv)), the restricted character is also $\sum_{\nu=\pi(\lambda)-\square} \chi^\nu$. By independence (Corollary 6.8(v)), λ and $\pi(\lambda)$ have the same predecessors, so $\pi(\lambda) = \lambda$ (Lemma 13.14). ∎

13.5 The Frobenius formula

Corollary 13.16 (The Frobenius formula). *For $\lambda, \mu \vdash n$, in any number $N \geq \ell(\lambda)$ of variables,*

$$\chi^\lambda(\mu) = [x^{\lambda+\delta}](a_\delta\, p_\mu).$$

Equivalently, $p_\mu = \sum_\lambda \chi^\lambda(\mu)\, s_\lambda$ for $N \geq n$.

Proof. $N \geq n$. By Lemma 13.11 and Theorem 13.15,

$$[x^{\lambda+\delta}](a_\delta\, p_\mu) = \langle \operatorname{ch}\chi^\lambda, p_\mu\rangle = \overline{\chi^\lambda(\mu)},$$

an integer, so real. The second form is the expansion of p_μ in the orthonormal basis s_λ.

Fewer variables. Let $\ell(\lambda) \leq N$. In $N+1$ variables the last entry of $\lambda+\delta$ is 0, so setting $x_{N+1} = 0$ keeps the coefficient. By Lemma 13.7(ii),

$$a_\delta(x_1,\dots,x_N,0) = x_1\cdots x_N\, a_\delta(x_1,\dots,x_N),$$

with the staircase of length N on the right; the factor $x_1\cdots x_N$ lowers the first N entries of $\lambda+\delta$ by 1, to the $\lambda+\delta$ of N variables. So the coefficient does not change from $N+1$ to N, and the formula descends from $N \geq n$. ∎

13.6 Worked example

$n = 3$. With χ_E from Example 6.4, and the fixed points of $M^{(2,1)} \cong \mathbb{C}^3$,

μ	(1^3)	$(2,1)$	(3)
z_μ	6	2	3
χ_E	2	0	-1
$\chi_{M^{(2,1)}}$	3	1	0

So

$$\operatorname{ch}(\chi_E) = \frac{2}{6}p_1^3 - \frac{1}{3}p_3 = \frac{1}{3}(p_1^3 - p_3).$$

As $E = V^{(2,1)}$, this is $s_{(2,1)}$. And

$$\operatorname{ch}(\chi_{M^{(2,1)}}) = \frac{1}{2}p_1^3 + \frac{1}{2}p_2p_1 = h_2h_1,$$

as, by Theorem 13.5(ii),

$$h_2 = \frac{1}{2}(p_1^2 + p_2), \qquad h_3 = \frac{1}{6}(p_1^3 + 3p_2p_1 + 2p_3).$$

So

$$h_2h_1 - h_3 = \frac{1}{6}(2p_1^3 - 2p_3) = \operatorname{ch}(\chi_E).$$

Injectivity gives $\chi_E = \chi_{M^{(2,1)}} - 1_{S_3}$, recovering Lecture 5's splitting.

An entry of S_4's table. Use $N = 2$ and $\lambda+\delta = (3,2)$:

$$\begin{aligned} a_\delta\, p_{(3,1)} &= (x_1-x_2)(x_1^3+x_2^3)(x_1+x_2) \\ &= x_1^5 - x_1^3x_2^2 + x_1^2x_2^3 - x_2^5, \end{aligned}$$

so $\chi^{(2,2)}((3,1)) = -1$, matching § 6.6's table, since $V^{(2,2)} \cong \rho_E \circ \pi$ (§ 12.4).

Self-check. Compute $\chi^{(3,1)}((2,2))$ by the Frobenius formula. *Answer:* with $N = 2$, $\lambda+\delta = (4,1)$, and

$$\begin{aligned} (x_1-x_2)(x_1^2+x_2^2)^2 = x_1^5 - x_1^4x_2 + 2x_1^3x_2^2 \\ - 2x_1^2x_2^3 + x_1x_2^4 - x_2^5, \end{aligned}$$

so $\chi^{(3,1)}((2,2)) = -1$, as for χ_W on $(1\,2)(3\,4)$: $V^{(3,1)} \cong W$ (§ 12.4).

Summary.

- *The main results:* ch is an isometry, with restriction $\partial/\partial p_1$ (Theorem 13.5); the s_λ are orthonormal (Theorem 13.10); $\mathrm{ch}(\chi^\lambda) = s_\lambda$ (Theorem 13.15); the Frobenius formula (Corollary 13.16).
- *Handed on:* $\mathrm{ch}(\chi^\lambda) = s_\lambda$ and $p_\mu = \sum_\lambda \chi^\lambda(\mu)\, s_\lambda$ to Lecture 14.
- *Next:* Lecture 14 finds $s_\lambda(x_1, \ldots, x_d)$ as a character of GL_d.

References

Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed.: p_λ a basis, (2.12), and z_λ, p. 24; a_α and s_λ, (3.1), p. 40; Cauchy's identities, (4.1), p. 62, and (4.3), p. 63; dual bases and orthonormality, (4.6)–(4.8), pp. 63–64; Cauchy's determinant, §I.4 Ex. 6, pp. 67–68; the characteristic, (7.2)–(7.3), p. 113; $\mathrm{ch}(\chi^\lambda) = s_\lambda$ and the Frobenius formula, (7.6)–(7.8), p. 114.[1] Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: Cauchy's determinant, Prop. 4.2.3, p. 173; Cauchy's identity for s_λ, Lemma 4.3.2, p. 186; the characteristic, §4.3.3, pp. 189–190. Fulton and Harris, *Representation Theory*: the Frobenius formula, 4.10, pp. 48–49.[2]

[1] Macdonald's products are bilinear, agreeing with ours on real functions.
[2] They index classes by numbers of j-cycles, and write Δ for a_δ and l for $\lambda + \delta$.